

OligoTox-Thrombocytopenia — Methodology

Materials and methods · NIH/NCATS OligoTox Open Data Challenge, Phase 2

Nature of the dataset: in-silico curation of pre-existing, publicly reported data. The "materials and methods" below therefore describe source identification, extraction, harmonisation, grading, provenance and quality control — how the dataset was assembled and computationally processed.

1. Materials — the sources

The material of this dataset is **70 primary sources**, each retrieved in full and cited to an exact locus. Four classes, in descending order of redistribution freedom:

Class	Examples	Retrieval route	rows
Regulatory — public domain	FDA prescribing information; FDA multidiscipline, clinical and pharmacology/toxicology reviews (NDA 211172, 217779, 203568, 210922, 212194, 214012, 215515, 206488, 217225); EMA SmPCs and EPAR assessment reports (Waylivra, Tegsedi, Onpattro, Oxlumo, Amvuttra, Qalsody)	DailyMed SPL REST API (structured XML); accessdata.fda.gov (browser User-Agent required); ema.europa.eu PDFs parsed with PyMuPDF	748
Patents — public domain	Ionis panels: US 2023/0263822, US 11,535,849, US 2021/0355497, US 2015/0031747, US 10,927,377, US 10,100,310	Google Patents HTML (USPTO print PDFs are image-only); examples tables and formal sequence listings	(within above)
CC-BY literature	Sewing 2017 (PLOS); Valenzuela 2023 (MDPI); the ASO adverse-event meta-analysis	Europe PMC / NCBI full-text XML plus supplementary files	227
Other literature — summary statistics	Crooke 2017 pooled safety database; Lundberg Slingsby 2022; Flierl 2015; Jaax 2013; Zaslavsky 2021; clinical trial reports	as above; PDF render where XML is abstract-only	984

Sources were discovered by a parallel sweep across ten independent lanes and ranked by **extractable per-measurement content**, not topicality. Each had to be actually fetched before being listed, so the registry contains no citation that cannot be re-checked. Regulatory review documents proved the richest and least-mined: they carry per-patient nadir listings, antibody cross-tabulations and dose-banded incidence grids absent from the corresponding journal papers.

2. Oligonucleotide identity, chemical modification and purity

2.1 How oligo identity was established and characterised

Because compounds were **not synthesised for this work**, "characterisation of identity" means establishing, from the primary record, exactly which molecule each row refers to. Four methods were used, each leaving an auditable trail in `design_source`:

- Patent sequence listings.** Formal listings give sequence and declared length. Every sequence was accepted only after matching an *independently declared* length — the LENGTH field, the target start/stop span, or the gapmer-motif sum. Across one patent, 2,631 sequences passed this dual check with zero conflicts.
- Deterministic parse of regulatory chemical names.** An FDA label's IUPAC chemical name spells out every residue. Parsing olezarsen's yielded `AGCTTCTTGTCAGCTTTAT`, and the stated molecular formula `C296H419N710154P20S19Na20` independently fixes 19 phosphorothioate linkages across 20 residues — an orthogonal confirmation of both sequence length and backbone.
- WHO INN nomenclature.** INN lists spell out each residue longhand, permitting recovery by deterministic parse rather than transcription.
- Cross-dataset agreement.** Sequences were checked against a sister endpoint's records, themselves validated against INN nomenclature and patent listings: **zero conflicts** across all shared compounds. Two independent derivation paths reaching the same

string is evidence against transcription artefact.

Sequences are recorded for 194 of 259 oligos (74 %) and were never guessed. The remainder are proprietary, class-level aggregates, or development-code compounds with no published sequence; they carry `TBD`.

2.2 Location of chemical modifications

Modification location is captured at three levels of resolution, according to what each source printed: `modification_map` holds **per-residue notation exactly as printed** (e.g. `mG*mC*mG*mA*...`, where `*` marks a phosphorothioate linkage and a leading letter a 2'-modified residue); `gapmer_design` gives wing-gap-wing boundaries (e.g. `5-10-5_MOE`) for 171 compounds; and `sugar_modifications` plus `ps_count` record which chemistries are present and the backbone load (237 and 198 compounds respectively). Where a source printed residue-level notation it is preserved verbatim rather than normalised away, because `sequence_5to3` holds bases only and would otherwise discard the positional detail.

Two encoding hazards were met and are recorded so others avoid them. **Letter case is significant** — it distinguishes modified wings from the DNA gap — so case must be read from the PDF text layer, never judged from rendered pixels. And in one source **chemistry is encoded by font colour and PS/PO by underline**; colour was recoverable from text-layer span attributes, underline was not, so those records retain `TBD` rather than a guess.

2.3 Purity data — a structural limitation, stated plainly

The Challenge requires the dataset to contain "data on the purity and characterization of each" oligo. **Purity is not available for the great majority of compounds in this dataset, and no purity value has been inferred, estimated, or carried across from another compound or batch.**

The requirement is written for submitters who *synthesise* the oligonucleotides they test and therefore hold HPLC, LC-MS or capillary-electrophoresis records for every batch. This dataset curates **published** results for compounds made by third parties, and the source literature almost never reports purity for the material it tested — a regulatory review states a platelet count, not a chromatogram.

The response is threefold rather than cosmetic. (i) The columns `purity_pct` and `purity_method` exist and are populated wherever a source *does* state purity. (ii) The QC gate **reports coverage of purity, characterisation method, sequence and modification map on every run**, so the shortfall is a visible standing warning rather than something a reader must discover. (iii) The limitation is declared here and in the dataset documentation.

Where identity assurance is available it is recorded in §2.1: for a large share of compounds the identity is confirmed against a formal patent sequence listing or a regulatory chemical name with a matching molecular formula, which is a stronger identity check than a purity percentage alone. Closing the purity gap properly would require either wet-lab characterisation of the compounds or access to sponsors' certificates of analysis; neither is achievable by curation, and we do not represent otherwise.

3. Extraction methods

Extraction ran as a multi-agent pipeline over the verified source list, one agent per source, each returning rows conforming to the published schema. Values were transcribed only from text actually retrieved. Tables were parsed from JATS XML where available, from PDF text layers via PyMuPDF otherwise, and two bitmap-only tables were recovered by rendering the page region at 4× and reading it visually.

Three table-extraction hazards occurred and are documented because each produces plausible, well-formed, wrong data rather than an error:

- 1. Line-wrapped sequence cells truncate silently.** A naive `[ACGT]` -run regex returns a plausible short sequence when a cell wraps; one run produced 94 corrupted sequences this way. *Guard:* accept a sequence only against an independently declared length.
- 2. Chemistry encoding is invisible to a renderer (§2.2).** *Guard:* resolve from the text layer; leave `TBD` where it cannot be.
- 3. Header/row misalignment returns the wrong analyte.** In three of five tables in one patent the header row omits the leading row-label cell, so addressing a column by header name is off by one — returning **red-cell counts where platelet counts were intended in 35 of 55 rows**, with entirely plausible magnitudes from the correct study. *Guard:* right-alignment of the numeric block, a units-row assertion, and a magnitude check; all 55 values were then re-verified against an independent manual transcription.

The general rule adopted: **validate every value against something outside the parse** — a declared length, a units row, a second host, or a hand transcription. A parser that is merely self-consistent fails silently.

No-fabrication policy. Sequences and toxicity values are never invented or recalled from memory. A value is recorded only if it appears in retrieved text, and every row must name its exact locus; a row whose locus cannot be named is dropped rather than kept with a vague citation. TBD is a correct and expected outcome. Compounds without published platelet data are omitted, not padded with assumed zeros.

4. Harmonisation and grading

All categorical fields use controlled vocabularies published in the schema and enforced by the QC gate. Each measurement carries an ordinal **thrombocytopenia_grade (0–3)**, CTCAE-aligned for clinical rows with explicit in-vitro analogues so bench and trial evidence share one scale. The rubric is built around the endpoint's bimodal presentation — the common dose-dependent decline versus the rare immune-mediated severe form — so that grading does not average them together.

Three grading conventions are applied uniformly: an **incidence row is graded by the severity of the event**, not by how small the incidence is; an **explicit reported zero is grade 0** and its notes record which grade the band would otherwise have carried, so severe-band cells stay findable; and **bleeding is graded on attribution**, requiring haemorrhage attributable to thrombocytopenia for grade 3. Grades are flagged provisional pending subject-matter review.

Human and animal evidence are separated by a derived `subject_class` column computed from study type × species and independently re-derived by the QC gate, which fails the build on disagreement. Pooled multi-species findings are assigned to neither side rather than forced.

5. Quality control and verification

`scripts/qc_thrombo.py` gates every ingestion round and exits non-zero on failure: schema conformance against the published enums; referential integrity (zero orphans); primary-key uniqueness; range checks; **provenance completeness** — no row may have an empty or TBD source identifier, reference or locus; case-insensitive sequence validation (case encodes chemistry, so a case-sensitive check would reject correct rows); `subject_class` consistency; and standing warnings for control-arm rows carrying non-zero grades, abstract-only citations, and Phase 2 content coverage.

Adversarial verification. Extracted rows are passed to an independent agent instructed to *refute* them: fetch the cited source and confirm it contains the claimed value at the claimed locus, that the grade follows the rubric, and that the row concerns platelets. Verdicts are applied mechanically, so a rejected row cannot survive by oversight. Two blocks totalling 659 rows have completed — **628 confirmed, 31 corrected, 0 rejected** — one of them by recomputing every value from the paper's raw per-replicate workbook, where every mean, SD and n reproduced exactly. **The remaining rows are curated but unverified**, and the dataset must not be described as fully verified until that completes.

Two further independent checks: a **held-out known-answer test** (the inotersen label parsed and its figures recorded before any extraction ran; all six matched) and **cross-dataset sequence agreement** (zero conflicts).

6. Reproducibility

The pipeline is a single command, `scripts/ingest_thrombo.sh`: harvest → merge lanes in a deterministic precedence order → assemble → re-apply committed verification verdicts → QC gate → rebuild the derived view and generated documentation. Determinism was verified by building twice with inputs supplied in opposite orders and confirming byte-identical output; an earlier order-dependence in conflict resolution was found and fixed this way.

The **full curation record is committed** — every agent's returned rows, the verification verdicts, and the source sweep — so the published tables are reproducible from their inputs, not merely re-checkable against their citations. Rebuilding from that record alone reproduces both CSVs byte-for-byte. Verdicts are keyed to a natural key drawn from row content rather than to row IDs, so verification survives the renumbering that re-assembly causes. All tooling is open (Python 3, PyMuPDF, scikit-learn, standard CSV) and no step depends on a proprietary service.

